

Use Case Place Order

1. Use Case Code

UC001

2. Brief Description

The use case “Place Order” describes the interaction between the customer and the AIMS Software. It is invoked when the customer wishes to place an order after the Customer has finished selecting products.

3. Actor(s)

There is only 1 Actor interacting with this use case: Customer.

4. Precondition(s)

- Customer has added at least 1 product with a non-zero quantity in cart.

5. Basic Flow of Events

Step 1. Customer places order

Step 2. AIMS Software checks for stock availability

Step 3. AIMS Software displays the delivery form (see Table 2) and prompts the customer to set up delivery information

Step 4. Customer enters delivery information and submits

Step 5. AIMS Software checks for missing required fields and input validity

Step 6. AIMS Software calculates and displays the delivery fee

Step 7. AIMS Software displays and temporarily saves invoice information (see Table 3)

Step 8. Customer requests to pay for the order

Step 9. AIMS Software calls the use case “Pay Order”

Step 10. AIMS Software puts the order in pending state

Step 11. AIMS Software displays general order information (see Table 4) and transaction information (see Table 5) to screen

Step 12. AIMS Software sends invoice and payment transaction information to customer's email

Step 13. AIMS Software records payment transaction information and successfully paid order

6. Alternative Flows

No	Location	Condition	Action	Resume Location
1	Step 2	Stock is insufficient	AIMS software prompts the customer to update the cart and displays the available quantity for each affected product.	Use case ends
2	Step 5	Missing required fields or invalid inputs	AIMS Software prompts the customer to fill in missing fields or reenter	Step 4
3	Step 7	Customer requests to adjust the delivery method	AIMS Software displays the delivery form and prompts the customer to reenter	Step 4
4	Step 7	Customer requests to adjust product list	AIMS Software displays cart and prompts the customer to update the cart	Step 1
5	Any step	Customer requests to go back to a step numbered N	AIMS Software redirects to that step	Step number N
6	Any step	Customer exits the software or requests to cancel order	AIMS Software cancels the ordering process	Use case ends
7	Step 9	Order payment is not successful	AIMS Software displays notification of failed transaction	Use case ends

Table 1-Alternative flows of events for UC Place Order

7. Input data

No	Data Fields	Description	Mandatory	Valid Condition	Example
1	Customer name	Name of the customer/receiver	Yes		Nguyễn Văn A
2	Phone number	Contact number of the customer that will be used to contact the customer	Yes	- 10 digits - Starts with 0 - May contain separators such as '-', '.', or space in between digits	0123456789 0123.456.789 0123-456-789 0123 456 789 012.34.56.789 ...
3	Email	Contact email of the customer that will be used to send order notifications	Yes	- Must be of valid form with an "@" symbol in between	john26@ex.com

4	Province/City (Tỉnh/Thành phố)	A province/city in Viet Nam, selectable from a list	Yes		Hà Nội Nghệ An ...
5	Ward/Commune (Phường/Xã)	A ward/commune in that province/city, selectable from a list	Yes		Phường Hà Đông Xã An Khánh ...
6	Address	Detailed address of the customer attached to	Yes		199 Chiến Thắng, Triều Khúc
7	Delivery method	How the products should be delivered to the customer	Yes		Quick, Normal (default), ...

Table 2-Input data of Delivery Information

8. Output data

No	Data Fields	Description	Display Format	Example
1	Product type + title	The type and the title of the product	[Type] [Title]	Book Data Structures & Algorithms DVD Ice Age Collection ...
2	Quantity	The number of the corresponding product to be purchased	Positive integer	3
3	Price	Price of the product	Positive integer Includes comma for thousands	99,000
4	Amount	Total price for the product	Positive integer Includes comma for thousands	297,000
5	Total product price	Total amount of all products	Positive integer Includes comma for thousands	1,423,000
6	Total product price with VAT	Total product price + VAT	Positive integer Includes comma for thousands	1,585,300
7	Delivery fee		Positive integer Includes comma for thousands	50,000
8	Total amount	The amount to be paid	Positive integer Includes comma for thousands	1,635,300

Table 3-Output data of Invoice Information

No	Data Fields	Description	Display Format	Example
1	Customer name			Nguyễn Văn A
2	Phone number		10 digits starting with 0 with '.', '-', or space	0123456789 0123.456.789 0123-456-789 0123 456 789 012.34.56.789 ...
3	Province/City (Tỉnh/Thành phố)			Hà Nội Nghệ An ...
4	Ward/Commune (Phường/Xã)			Phường Hà Đông Xã An Khánh ...
5	Address			199 Chiến Thắng, Triều Khúc
6	Delivery instructions			
7	Total amount	The amount to be paid	Positive integer Includes comma for thousands	1,635,300

Table 4-Output data of Order Information

No	Data Fields	Description	Display Format	Example
1	Transaction ID			
2	Transaction Content			
3	Transaction Datetime		dd/mm/yyyy	15/03/2026

Table 5-Output data of Transaction Information

9. Postconditions

- A new order is created in the system and its status is set to “Pending”.
- The system records payment transaction information and the successfully paid order
- Customer receives email containing invoice and payment transaction information

If the use case ends prematurely or is unsuccessful due to payment failure, the ordering process is reset and nothing else happens.